

HSL-MIP: An Effective Hybrid Stacked Learning–Based Model for Myocardial Infarction Prediction

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Abstract—Myocardial infarction, commonly known as a heart attack, remains a major cause of death worldwide. Various machine learning methods have shown potential in early risk detection. However, many existing approaches place greater emphasis on overall accuracy than on sensitivity. In clinical screening, this may result in false negatives where individuals at risk are not identified. To address this issue, this study proposes HSL-MIP, a Hybrid Stacked Learning model developed with an emphasis on improving recall. This framework combines boosting-based and bagging-based tree models to enhance the bias-variance trade-off control for myocardial infarction risk prediction. The Multi-Layer Perceptron (MLP) meta-classifier further strengthens the architecture by modeling high-order non-linear correlations among base learner predictions. The model was evaluated on a publicly available dataset of 1,190 patient records. The proposed HSL-MIP achieved a Recall of 96.95% and an Accuracy of 95.80%, significantly reducing false negative cases to just 4 cases on the held-out test set. These results demonstrate that HSL-MIP offers a more clinically safe and robust alternative to standalone algorithms for the early detection of myocardial infarction.

Keywords—Hybrid Stacking Ensemble Model, Myocardial Infarction, Heart Attack Risk Prediction, Recall Optimization, Hyperparameter Tuning

I. INTRODUCTION

The death toll from Cardiovascular Diseases (CVDs) has risen to around 20 million as of 2022, with mortality trends continuing upward. CVDs cause 800,000 US heart attacks annually and 75% of CVD deaths in lower- and middle-income countries; in India, over 27% of deaths are CVD-related, nearly half occurring between ages 40–69 [1]. A heart attack occurs when blood flow to the heart is suddenly blocked, risking tissue damage if not promptly restored [2]. Risk is driven by elevated blood pressure, fasting blood sugar, and abnormal ECG values, compounded by smoking, high-cholesterol diets, hypertension, sedentary habits [3], stress, and poor diet [4] – all factors that can be monitored to predict heart attack risk [5].

Traditional diagnostics such as ECG, echocardiograms, and angiography detect cardiac abnormalities [6], but are limited in

early-stage detection, require specialized equipment, and can be time-consuming or inaccurate, motivating a noninvasive data-driven approach for early myocardial infarction detection.

Early prediction is challenged by complex, non-linear interactions among clinical risk factors. Machine learning offers a powerful alternative to traditional statistics for uncovering such patterns in large patient datasets. However, existing studies often prioritize overall accuracy over sensitivity (recall), leading to dangerous false negatives.

This study addresses these gaps through HSL-MIP, which uses a stacking of multiple models to improve the prediction reliability while mitigating overfitting. We evaluate different supervised algorithms to identify the best fit for our dataset. We also address the heavy reliance on single-source datasets by using a mixed Kaggle dataset compiled from five widely used heart attack risk datasets, and the heavy reliance on standalone algorithms that often lack the high recall required for clinical validation, since high recall reduces the chance of incorrectly labeling a patient as a safe case. Our model achieves a high recall of 96.95%, supporting its use in clinical workflows to help clinicians make faster, more accurate diagnoses.

II. LITERATURE SURVEY

Considerable advances have been made in ML-based CVD prediction. Miah et al. [7], Biswas et al. [8], Alshraideh et al. [9], and Saboor et al. [10] reported strong accuracies (90–96%) with XGBoost, Random Forest, and SVM, but on small, regional, or single-institution datasets, limiting generalizability. Shijitha et al. [11] used a larger Kaggle dataset (>4,000 records) with Logistic Regression (88% accuracy), though linear separability assumptions limit its ability to capture MI’s non-linear patterns.

Feng et al. [12] and Yang [13] used XGBoost for its non-linear capability, but relied solely on a single boosting learner. Khan et al. [14] found LightGBM best (84.60%) on a 600-sample dataset, while Sumwiza et al. [15] achieved 99% accuracy

TABLE I
HEART DISEASE DATASET DESCRIPTION

S.No.	Attribute	Code given	Unit	Data type	Description
1	age	Age	in years	Numeric	
2	sex	Sex	1, 0	Binary	1 = male; 0 = female
3	chest pain type	chest pain type	1, 2, 3, 4	Categorical	Value 1: typical angina Value 2: atypical angina Value 3: non-anginal pain Value 4: asymptomatic
4	resting blood pressure	resting bp s	in mm Hg	Numeric	
5	serum cholesterol	cholesterol	in mg/dl	Numeric	
6	fasting blood sugar	fasting blood sugar	1, 0	Binary	1 = true (if fasting blood sugar > 120 mg/dl) 0 = false
7	resting electrocardiogram results	resting ecg	0, 1, 2	Categorical	Value 0: normal Value 1: having ST-T wave abnormality Value 2: showing probable or definite left ventricular hypertrophy
8	maximum heart rate achieved	max heart rate	60-202	Numeric	
9	exercise induced angina	exercise angina	0, 1	Binary	1 = yes; 0 = no
10	oldpeak=ST	oldpeak	depression	Numeric	
11	the slope of the peak exercise ST segment	ST slope	1, 2, 3	Categorical	Value 1: upsloping Value 2: flat Value 3: downsloping
12	class	target	0, 1	Binary	1 = heart disease, 0 = Normal

with Random Forest; both remain limited by single-learner tree architectures lacking heterogeneous ensemble integration.

Other research works target more sophisticated pipelines: Waqar et al. [16] combined deep learning with SMOTE (96% accuracy); Bharti et al. [17] combined ML with an ANN (94.2%); Chauhan et al. [18] achieved 99.02% on a 1,025-record dataset with tuned Gradient Boosting, where the most influential feature was identified as chest pain. Cheng et al. [19] utilized a 132,000 record dataset from the Taiwan Biobank with Gradient Boosting, while Karna et al. [20] reviewed the field and highlighted hybrid models as a future direction. Despite this progress, few studies explore hybrid or stacking ensembles, and fewer still prioritize recall, which is vital to minimizing false negatives that carry fatal clinical consequences.

This study attempts to address these gaps by using a combined collection of five widely recognized datasets and proposing HSL-MIP, a hybrid stacking model which is trained by considering not only the overall accuracy but also giving more importance towards achieving a higher recall score, since identifying the positive cases correctly can support more reliable clinical decision-making.

III. METHODOLOGY

A. Data and Dataset Description

The dataset for this study was taken from Kaggle [21]. A compilation of structured clinical data was created by merging five widely recognized datasets which include Cleveland, Hungarian, Switzerland, Long Beach VA, and the Statlog (Heart) dataset. Together, these sources provided 1,190 patient records. Each record contains 11 attributes and one target variable. The dataset has a balanced target class, with 629 heart attack cases and 561 healthy cases.

The attributes include numerical, binary (0/1), and categorical (nominal) values. Their descriptions are presented in Table I.

B. Exploratory Data Analysis (EDA) & Data Pre-processing

To ensure the integrity of the data prior to model development, a systematic data exploration and pre-processing strategy was

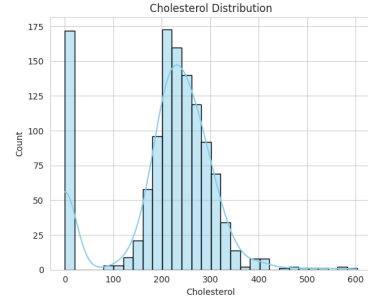


Fig. 1. Cholesterol distribution.

applied. An initial analysis of the dataset revealed the absence of explicit missing values. The 'cholesterol' feature had 172 zero-valued instances as shown in Figure 1. These values were treated as invalid observations and replaced them with a NaN value, and after splitting the train and test, the mean of the non-zero cholesterol measurements calculated from the training partition (245.9067 mg/dl) was assigned and this training-derived mean was applied to both the training and test sets.

A feature correlation heatmap was also analyzed (Figure 2). It showed ST Slope and exercise angina were positively correlated with the target. Max heart rate and cholesterol were negatively correlated. It is observed that correlations were moderate with no significant multicollinearity, indicating balanced feature contributions and suitability of the dataset for ML modeling.

C. Feature Engineering

- **Train-Test Split:** To prevent data leakage, the dataset is first partitioned before any feature transformation. The dataset $\mathcal{D}$ was partitioned into training (80%, 952 samples) and testing (20%, 238 samples) sets.
- **Standard Scaling:** StandardScaler was fitted exclusively on the training set and applied to both subsets, ensuring all features contribute equally during learning while preserving test-set integrity.

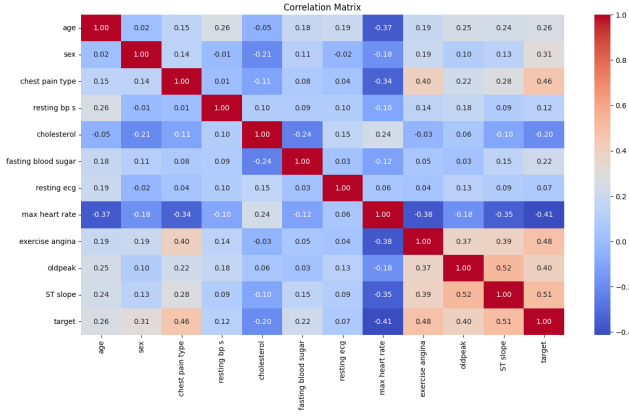


Fig. 2. Correlation matrix heatmap depicting the linear correlation between features and the target variable.

- **Hyperparameter Tuning:** GridSearchCV with 5-fold cross-validation was used to optimize all evaluated models using accuracy as the scoring metric. The selected HSL-MIP configuration comprised XGBoost with $n_{\text{estimators}} = 250$, $\text{max_depth} = 7$, $\text{learning_rate} = 0.05$, $\text{subsample} = 0.7$, and $\text{colsample_bytree} = 0.7$, combined with Random Forest using $n_{\text{estimators}} = 200$, $\text{max_depth} = \text{None}$, $\text{max_features} = \text{sqrt}$, $\text{min_samples_leaf} = 1$, and $\text{min_samples_split} = 2$.

D. Proposed HSL-MIP Model

To improve predictive performance and reduce missed positive cases, a hybrid HSL-MIP model was developed by combining two highly effective classifiers using a stacked ensemble approach: XGBoost and Random Forest. The models were selected based on their complementary learning mechanisms, with XGBoost employing gradient boosting and Random Forest utilizing bagging, thereby providing robust decision-making across varied data patterns. The stacking framework was implemented using 5-fold stratified cross-validation on the training set to construct the meta-features. In each fold, the base learners were trained on a subset of the training data and generated out-of-fold (OOF) probability predictions for the validation data. These probability outputs were directly used to form the meta-feature space.

During model inference, predictions from each base model on the test set across all folds were averaged and used as meta features for the final prediction. For the meta-learner, an MLP classifier was employed, consisting of a single hidden layer with 10 neurons, and trained for a maximum of 1000 iterations. The MLP was selected for its ability to capture non-linear relationships between the base model confidence scores. The overall architecture of the proposed HSL-MIP model is illustrated in Figure 3.

E. Mathematical Formulation of HSL-MIP

Let the data set be defined as $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, where $\mathbf{x}_i \in \mathbb{R}^d$ is the vector of d clinical features for patient i , and $y_i \in \{0, 1\}$

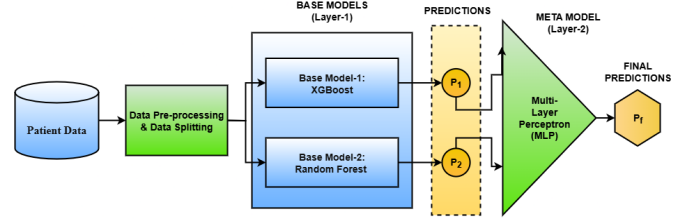


Fig. 3. Overall architecture of the proposed HSL-MIP model, which utilizes XGBoost and Random Forest as base learners and an MLP as the meta-classifier.

is the target (0: Normal, 1: Myocardial Infarction). HSL-MIP employs a two-tier stacking architecture.

1) *Level-0: Heterogeneous Base Learners:* We use two base learners, f_{xgb} (XGBoost) and f_{rf} (Random Forest), chosen for their complementary error profiles (bias-variance trade-off).

a) *Base Learner 1: XGBoost (f_{xgb}):* XGBoost is an additive ensemble of gradient-boosted decision trees. At iteration t , it minimizes:

$$\mathcal{L}^{(t)} = \sum_{i=1}^N l(y_i, \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)) + \Omega(f_t) \quad (1)$$

where l is the logistic loss function and $\Omega(f_t) = \gamma T + \frac{1}{2} \lambda \|w\|^2$ is the regularization term controlling tree complexity (T) and leaf weights (w).

b) *Base Learner 2: Random Forest (f_{rf}):* Random Forest aggregates B decision trees $\{T_1, \dots, T_B\}$ trained on bootstrap samples, with prediction via probabilistic averaging:

$$f_{rf}(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^B P_b(y = 1|\mathbf{x}) \quad (2)$$

2) *Level-1: Meta-Feature Generation:* To prevent data leakage, meta-features are generated via K -fold cross-validation ($K = 5$): $\mathcal{D}$ is partitioned into disjoint folds $\{\mathcal{F}_1, \dots, \mathcal{F}_K\}$, and for each fold k the base learners are trained on $\mathcal{D} \setminus \mathcal{F}_k$ and predict on $\mathbf{x}_j \in \mathcal{F}_k$. The meta-feature vector $\mathbf{z}_i$ is:

$$\mathbf{z}_i = \begin{bmatrix} \hat{y}_{i,xgb} \\ \hat{y}_{i,rf} \end{bmatrix} \in \mathbb{R}^2 \quad (3)$$

where $\hat{y}_{i,\cdot}$ is the out-of-fold probability estimate.

3) *Level-2: MLP Meta-Learner:* The meta-classifier is an MLP $\mathcal{H}$ mapping $\mathbf{z}_i$ to the final risk probability, with a single hidden layer of 10 neurons:

$$\mathbf{h}_i = \text{ReLU}(\mathbf{W}_h \mathbf{z}_i + \mathbf{b}_h) \quad (4)$$

followed by a sigmoid output:

$$\hat{y}_{final} = \frac{1}{1 + e^{-(\mathbf{w}_{out}^T \mathbf{h}_i + b_{out})}} \quad (5)$$

The MLP meta-learner is trained on the generated meta-dataset to minimize the classification log-loss.

F. State-of-the-Art Models

The models in this study include several classification algorithms. They are categorized by algorithm type in Table II.

TABLE II
STATE-OF-THE-ART ML MODELS USED AND THEIR
CORRESPONDING ALGORITHM TYPES

Model	Algorithm Type
XGBoost + Random Forest → MLP (HSL-MIP)	Stacking Ensemble
XGBoost + Random Forest → Logistic Regression	Stacking Ensemble
XGBoost + CatBoost → LR	Stacking Ensemble
Random Forest + XGBoost (Soft Voting)	Ensemble Voting
XGBoost	Boosting
Random Forest	Bagging
K-Nearest Neighbors (KNN)	Distance-Based
Gradient Boosting (GBM)	Boosting
CatBoost	Boosting
LightGBM	Boosting
Extra Trees	Bagging
Artificial Neural Network (ANN)	Neural Network
Support Vector Classifier (SVC)	Margin-Based
Decision Tree	Tree-Based
Logistic Regression (LR)	Linear Model
Naive Bayes (NB)	Probabilistic

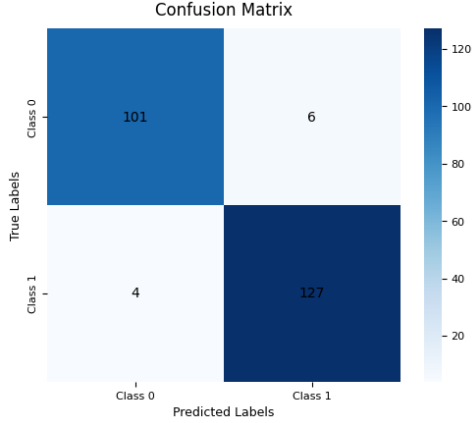


Fig. 4. Confusion matrix of the proposed hybrid model (XGBoost + Random Forest → MLP) showing $\begin{bmatrix} 101 & 6 \\ 4 & 127 \end{bmatrix}$. Only four false negatives were recorded.

IV. RESULTS AND DISCUSSION

The HSL-MIP model, which combines XGBoost and Random Forest as base learners with a Multi-Layer Perceptron (MLP) as the meta-classifier, delivered the best overall performance out of all models tested. It achieved the highest accuracy (95.80%), recall (96.95%), and F1-score (96.21%). These figures, and all other per-model results in Table III, come from a single held-out test set (238 samples, 20% of the cohort); the repeated cross-validation reported in Section IV-A is used only to compare candidate meta-learners and does not alter these test metrics. The confusion matrix of the proposed HSL-MIP model is shown in Figure 4, where only four false negatives were observed, highlighting its strong recall performance.

This high recall reflects HSL-MIP’s effectiveness at minimizing false negatives for recall-oriented clinical screening. Figure 5, illustrates the comparative performance of the top-performing models.

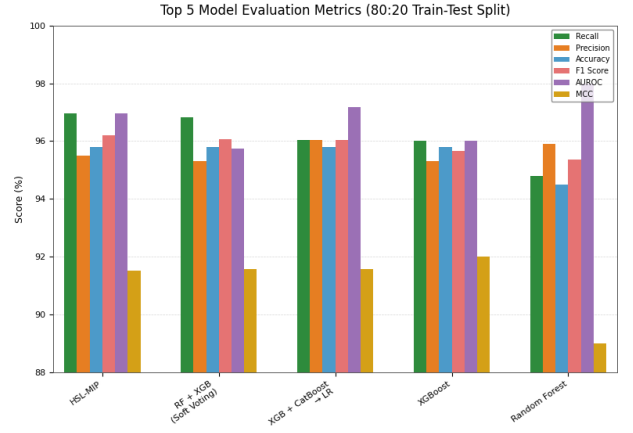


Fig. 5. Evaluation metrics for the top five performing models. The proposed hybrid model shows superior recall and overall performance. AUC values are scaled to match percentage scale.

A. Meta-Learner Comparison and Statistical Validation

To justify the selection of the MLP meta-learner, we compared it against Logistic Regression (LR) as an alternative meta-classifier under identical conditions, the same XGBoost and Random Forest meta-features using Repeated Stratified K -Fold Cross-Validation (10 folds $\times$ 5 repeats, 50 folds total). Table IV summarizes the mean performance with 95% bootstrap confidence intervals (10,000 resamples).

The MLP produced the higher mean recall of the two candidates (0.9577 vs. 0.9548), the metric most directly tied to the clinical objective of HSL-MIP: minimizing missed diagnoses. LR in turn produced a marginally higher mean ROC-AUC (0.9689 vs. 0.9663) and precision (0.9337 vs. 0.9327); accuracy, F1, and MCC were closely matched between the two candidates.

Because the 50 fold-level results produced by repeated K -fold cross-validation are not independent samples, the same patient records recur across multiple train/validation partitions between repeats. A standard paired t-test understates the true variance of the estimated performance difference. We therefore report the corrected resampled paired t-test (Nadeau and Bengio, 2003; Bouckaert and Frank, 2004), which inflates the variance estimate by a term reflecting the ratio of the validation-set size to the training-set size, yielding a more conservative and realistic significance estimate (Table V).

Given that recall is the primary design objective of HSL-MIP, and that the MLP’s mean recall was higher across the evaluation (Table IV), and that accuracy, F1, and MCC remained statistically comparable between the two candidates, the MLP was retained as the final meta-classifier.

Figure 6. presents a SHAP summary plot for the MLP meta-learner, showing the relative contribution of the XGBoost and Random Forest meta-features to the predicted risk probability.

B. Computational Complexity Analysis

The theoretical complexity of each component is as follows: XGBoost has $O(NKdD)$ complexity, where N is the number of

TABLE III
QUANTITATIVE ANALYSIS OF THE PROPOSED HSL-MIP MODEL COMPARED WITH THE STATE-OF-THE-ART ML MODELS (STANDALONE AND HYBRID)

Model Name	Performance Metrics					
	Acc.	Prec.	Recall	F1 Score	AUROC	MCC
Naive Bayes	81.93	82.17	84.13	83.14	0.9109	0.6371
Logistic Regression	87.82	88.19	88.89	88.54	0.8775	0.7554
ANN	87.82	91.45	84.92	88.07	0.9354	0.7587
Decision Tree	90.34	93.28	88.10	90.61	0.9048	0.8081
SVM	90.76	91.94	90.48	91.20	0.9189	0.8148
Extra Trees	91.60	94.17	89.68	91.87	0.9692	0.8329
LightGBM	92.02	95.73	88.89	92.18	0.9630	0.8429
CatBoost	93.28	94.35	92.86	93.60	0.9705	0.8653
GBM	94.54	95.93	93.65	94.78	0.9459	0.8908
KNN	94.54	95.20	94.44	94.82	0.9690	0.8905
Random Forest	94.54	95.93	93.65	94.78	0.9793	0.8908
XGB + CatBoost → LR	94.54	94.03	96.18	95.09	0.9674	0.8896
XGBoost	94.96	95.97	94.44	95.20	0.9499	0.8990
RF + XGB (Soft Voting)	95.38	95.45	96.18	95.82	0.9733	0.9066
HSL-MIP	95.80	95.49	96.95	96.21	0.9695	0.9151

TABLE IV
PERFORMANCE COMPARISON: MLP VS. LOGISTIC REGRESSION META-LEARNERS

Metric	LR Mean ± SD	LR 95% CI	MLP Mean ± SD	MLP 95% CI
Accuracy	0.9398 ± 0.0222	0.9336–0.9459	0.9407 ± 0.0234	0.9339–0.9471
Precision	0.9337 ± 0.0275	0.9259–0.9412	0.9327 ± 0.0279	0.9248–0.9403
Recall	0.9548 ± 0.0299	0.9464–0.9628	0.9577 ± 0.0305	0.9491–0.9659
F1 Score	0.9437 ± 0.0208	0.9378–0.9493	0.9446 ± 0.0219	0.9384–0.9506
MCC	0.8803 ± 0.0440	0.8679–0.8923	0.8820 ± 0.0466	0.8687–0.8947
ROC AUC	0.9689 ± 0.0145	0.9648–0.9729	0.9663 ± 0.0156	0.9618–0.9705

TABLE V
CORRECTED RESAMPLED PAIRED T-TEST RESULTS: LR VS. MLP META-LEARNERS

Metric	t-statistic	p-value	Significance
Accuracy	-0.5077	0.6140	Not significant
Precision	0.7166	0.4770	Not significant
Recall	-1.0322	0.3071	Not significant
F1 Score	-0.5674	0.5730	Not significant
MCC	-0.5160	0.6082	Not significant
ROC AUC	2.3315	0.0239	LR higher

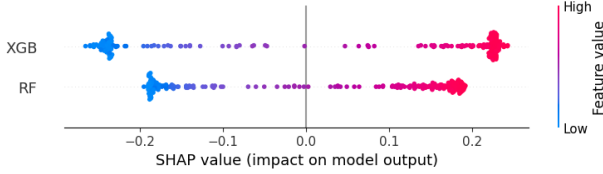


Fig. 6. SHAP summary plot for the MLP meta-learner, illustrating the relative contribution of the XGBoost and Random Forest meta-features to the predicted risk probability.

TABLE VI
COMPUTATIONAL COMPLEXITY ANALYSIS OF HSL-MIP COMPONENTS

Component	Approximate Complexity
XGBoost (fit)	$O(NKdD)$
Random Forest (fit)	$O(MNd \log N)$
Base-model prediction	$O(KdD + Md \log N)$
MLP meta-learner (fit)	$O(ENh)$
5-fold stacking	$O(F[NKdD + MNd \log N])$
Overall HSL-MIP	$O(F[NKdD + MNd \log N] + ENh)$

samples, K is the number of trees, d is the number of features, and D is the maximum tree depth. Random Forest has $O(MNd \log N)$ complexity, where M is the number of trees, N is the number of samples, and d is the number of features. The MLP meta-learner has $O(ENh)$ complexity, where E is the number of epochs, N is the number of samples, and h is the number of hidden neurons.

Table VI summarizes the computational complexity of the HSL-MIP components, while the measured execution times were obtained from the 5-fold out-of-fold procedure used to construct the stacking meta-features on the training partition (952 samples, 80% of the 1,190-record dataset). The Random Forest base learner required 7.8426 seconds of total fit time and 0.4035 seconds of total prediction time across the five folds reaching a total of 8.2462 seconds, demonstrating efficient scalability for the dataset size. These results confirm that the proposed HSL-MIP architecture maintains reasonable computational requirements suitable for further clinical evaluation.

V. CONCLUSION

This study showcased the effectiveness of hybrid stacked models in predicting the risk of myocardial infarction. A range of ensemble learners and classical classifiers were examined during the evaluation. Among the individual models, XGBoost produced the highest accuracy, whereas Random Forest achieved the highest AUROC. The results also showed that data pre-processing, feature scaling, and hyperparameter tuning played an important role in the performance of the models.

Based on the performance of the individual classifiers, a hybrid stacked learner (HSL-MIP) was developed using XGBoost and Random Forest as the base learners and an MLP as the meta-classifier. Across all the models tested, the hybrid model obtained the highest recall (96.95%) and Matthews Correlation Coefficient (0.9151). It also achieved 95.80% accuracy, a 96.21% F1-score, and an AUROC of 0.9695. The high recall is important in this case because missing patients

who may be at risk of myocardial infarction can have serious consequences in clinical practice.

There are some limitations to the study. The model was evaluated using a consolidated dataset containing 1,190 records obtained from five merged sources. As a result, the dataset may not fully represent variations found across different clinical populations or geographic regions. The performance of HSL-MIP should therefore be examined using independent external datasets. Testing the model separately across different data sources may also provide a better understanding of how well it performs when applied beyond the data used in this study.

There are several areas that can be explored in future work. Different machine learning approaches, including deep learning models, could be tested to examine whether they can identify more complex patterns in clinical data. Further feature engineering may also be useful. This can involve automated feature selection, the construction of features based on domain knowledge, and the extraction of information from time-series data. In addition, including more detailed patient medical histories and structured clinical information may help improve the prediction process and support earlier and more personalized assessment of myocardial infarction risk.

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